

FooFoo — Recommendation Engine Persona Validation

Report generated: 2026-05-27 | Run: 50 personas | Source: GitHub Actions · Persona Validation workflow

Key finding: The recommendation engine code is now working correctly (auth bugs and schema cache bugs fixed). All 50 failures are **data gaps** — the dish database is missing the data the engine needs to generate personalised plans. No code changes are required to fix these.

Executive Summary

Issue	Personas Affected	Count	Severity	Type
Dishes not tagged with meal slots (breakfast / lunch / dinner)	P001 – P043	43	HIGH	Data gap — dishes table
No vegan dishes exist in the database	P044 – P050	7	HIGH	Data gap — dishes table
Cuisine codes in persona definitions missing from cuisines_master	20+ personas	~25 gaps	MEDIUM	Data gap — cuisines_master table

Issue 1 — Eligible Dish Pool = 0 (P001–P043, 43 personas)

Affects all non-vegan personas. The dish query succeeds and returns rows, but after the RE scoring filters (diet type + meal slot compatibility), zero dishes are eligible for any slot.

Question	Answer
What happened	Plan generator ran, fetched dishes, applied scoring filters — but 0 dishes matched the breakfast / lunch / dinner slot requirements. A fallback assigned the first available dish, so the plan was saved, but the carousel was empty.
Why	The meal_types column on the dishes table is either NULL or contains values that don't match what the RE scoring engine expects (e.g. "breakfast", "lunch", "dinner", "all"). Without this tag, no dish passes the slot filter.
What the user sees	App shows a plan with dishes, but they may not be contextually appropriate (breakfast dishes at dinner, etc.).
How to fix	Populate the meal_types column for all 818+ dishes in the dishes table. Use UPDATE dishes SET meal_types = ARRAY['lunch','dinner'] WHERE ... style SQL grouped by dish type (e.g. rice, curry, dal = lunch/dinner; idli, poha, upma = breakfast; snacks = all).
Priority	HIGH — fix first — single change unblocks 43 of 50 personas

Issue 2 — No Vegan Dishes in Database (P044–P050, 7 personas)

All 7 vegan personas hard-fail. The dish query itself returns 0 rows because no dishes have `diet_type = 'vegan'`.

Question	Answer
What happened	Edge function returned HTTP 422 "No eligible dishes match the diet/allergen filters" — plan not generated at all.
Why	The dishes table contains no rows where <code>diet_type = 'vegan'</code> . Vegan is a sub-category of vegetarian — many veg dishes are technically vegan but are not tagged as such.
How to fix	Two options: (A) Add dishes with <code>diet_type = 'vegan'</code> , or (B) run <code>UPDATE dishes SET diet_type = 'vegan' WHERE ...</code> for dishes that contain no dairy/egg ingredients. Option B can be automated via the <code>derive-dish-attributes</code> Edge Function.
Priority	HIGH — vegan is a supported user diet type in the app

Issue 3 — Cuisine Codes Missing from `cuisines_master` (20+ personas)

Persona preference definitions reference cuisine codes that don't exist in the `cuisines_master` table. The RE cannot apply cuisine boosts for these preferences.

Missing Cuisine Codes by Category

Category	Missing Codes	Personas Affected
Jain variants	<code>jain</code> · <code>rajasthani_jain</code> · <code>north_indian_jain</code> · <code>maharashtrian_jain</code> · <code>street_food_jain</code> · <code>gujarati_jain</code>	P028–P035
Vegan variants	<code>north_indian_vegan</code> · <code>continental_vegan</code> · <code>indo_chinese_vegan</code> · <code>street_food_vegan</code> · <code>bengali_vegan</code> · <code>tamil_vegan</code> · <code>kerala_vegan</code> · <code>maharashtrian_vegan</code>	P044–P050
Regional South Indian	<code>tamil_brahmin</code> · <code>telugu</code> · <code>kerala_vegan</code> · <code>south_indian_jain</code>	P002, P011, P024, P034
Regional Bengali	<code>bengali_veg</code> · <code>bengali_vegan</code>	P009, P045, P049
Rajasthani / Marwari	<code>marwari</code> · <code>rajasthani_jain</code>	P014, P029, P033, P035
North East India	<code>north_east</code>	P027, P040
Coastal / Goan	<code>seafood</code> · <code>coastal</code> · <code>konkan</code> · <code>goan_vegetarian</code>	P017, P022, P038
Central India	<code>madhya_pradeshi</code>	P012
Mughlai / Awadhi variants	<code>mughlai_veg_egg</code> · <code>awadhi_veg</code>	P042
Fusion / Other	<code>persian_influenced</code> · <code>all_others</code>	P021, P030, P033, P035

Question	Answer
Impact	Cuisine boost doesn't apply for missing cuisines — but this alone doesn't cause plan failures. It means plan quality is lower for these personas (no personalisation on their preferred cuisines).
How to fix	Add the missing cuisine codes to the cuisines_master table, then tag existing dishes with the appropriate new cuisine IDs. Alternatively, map these codes to existing parent cuisines (e.g. tamil_brahmin → south_indian).
Priority	MEDIUM — does not block plan generation; improves personalisation quality

Recommended Fix Order

#	Action	Where	Impact	Effort
1	Populate meal_types column on all dishes	dishes table — bulk SQL UPDATE	Unblocks 43 personas	Low — SQL only
2	Tag veg dishes that are vegan as diet_type = 'vegan'	dishes table — SQL UPDATE or derive-dish-attributes Edge Fn	Unblocks 7 vegan personas	Low — SQL or Edge Fn
3	Add missing regional / sub-cuisine codes	cuisines_master table + re-tag dishes	Improves RE personalisation quality	Medium — data work
4	Re-run Persona Validation workflow	GitHub Actions	Confirms fixes green	1 click

What IS Working (not broken)

Component	Status
Edge function authentication (user JWT path)	✅ Fixed — was "Invalid or expired token", now resolved
PostgREST dishes query (cuisines_master embed removed)	✅ Fixed — was "relationship not found in schema cache", now resolved
RE scoring pipeline (diet filter, allergen filter, cuisine boost, weather boost)	✅ Working — filtering correctly, issue is data not code
Plan saving to DB (planner + planner_carousel tables)	✅ Working — fallback saves plan even when pool is 0
Test user creation + cleanup	✅ Working — 50 users created and deleted cleanly